Confluent Cloud Lab

Overview

Objectives

By the end of this lab, learners will be able to:

- Navigate **environments** and **Kafka clusters** in Confluent Cloud.
- Create and inspect **topics** and **partitions** using the **Topics** UI.
- Produce and browse **messages** with the **Message Browser**.
- Understand **keys, values, offsets, partitions, and timestamps** at the message level.
- Use a managed **Datagen connector** to simulate producers and observe **streams of events**.
- Locate and interpret **consumer group lag** and consumer activity in the UI.

Prerequisites

- A Confluent Cloud account (free trial or paid).
- At least one **environment** and a **Kafka cluster** (Basic or Standard is fine). If needed, follow Confluent's "Quick Start for Confluent Cloud" to create an environment and a cluster before this lab.

Part 1 – Explore Environments and Clusters

Goal: Map **Kafka concepts** (cluster, bootstrap, endpoint) to what you see in Confluent Cloud.

1. **Log in to Confluent Cloud**
    - Go to https://confluent.cloud and sign in.
    - On the left, notice the **environment selector** (usually showing a name like default or one you created).
2. **Select or Create an Environment**
    - If multiple environments exist, choose your target environment from the dropdown at the top.
    - If none exist, click **Add environment** and create one (e.g., training-env).

3. **Open a Kafka Cluster**
    - Within the environment, select an existing **Kafka cluster**, or create a new **Basic** cluster from the **Clusters** page (give it a name like kafka-concepts-lab).
    - Once provisioned, click into the cluster.
    - Note:
        - **Bootstrap endpoint** (hostname:port).
        - **Cloud & region** (e.g., AWS / us-west-2).
        - **Cluster type** (Basic/Standard/Dedicated).

Part 2 – Topics and Partitions in the UI

Goal: Use the **Topics** page to connect the ideas of topics, partitions, replication, and configs.

1. **Navigate to Topics**
    - In the left navigation of the cluster, click **Topics**.
    - Review any existing topics (system topics may be visible depending on your plan).
2. **Create a New Topic**
    - Click **Create topic** (or **Add a topic**, depending on the UI state).
    - Set:
        - **Topic name**: orders-concepts-lab
        - **Partitions**: 3
        - Keep replication to the default for this cluster (Confluent Cloud manages RF based on cluster type).
    - Click **Create with defaults** (you can mention that **Customize settings** allows changing retention, cleanup policy, etc.).
3. **Inspect the Topic Detail Page**
    - Click on orders-concepts-lab to open its details page.
    - Observe:
        - **Number of partitions**.

        - **Cleanup policy**, **retention period**, and other configs (view under “Configuration” tab).
        - Visual partition layout (per-partition metrics may appear once data flows).

Part 3 – Message Browser: Offsets, Keys, and Values

Goal: See **actual Kafka records** (offset, partition, timestamp, key, value) via the **Message Browser**.

1. **Open Message Browser for orders-concepts-lab**
    - On the orders-concepts-lab topic page, click the **Messages** or **Message browser** tab (name can vary slightly).
    - If no data exists yet, the pane will be empty and may prompt you to produce a message.
2. **Produce a Test Message via UI**
    - Click **Produce a message** in the message browser.
    - In the dialog:
        - Leave **Key** empty for now (or add "order-1" as a simple string key).
        - In **Value**, add JSON like:

{ "orderId": 1, "customer": "Alice", "amount": 99.9 }

    - Click **Produce**.
3. **Inspect the Message**
    - In the messages table, locate your new record.
    - Note:
        - **Partition** number assigned.
        - **Offset** within that partition.
        - **Timestamp** (ingestion time).
        - **Key** and **Value**.

4. **Seek by Offset/Time (Optional)**
    - Use the **Seek** control in message browser to jump to:
        - **Beginning** of topic.
        - A specific **offset**.
        - A particular **timestamp**.

Part 4 – Simulate Producers with the Datagen Connector

Goal: Use a **managed connector** to continuously produce data into a topic, showing producers and streams without writing code.

1. **Create a New Topic for Datagen**
    - Go to **Topics** → **Create topic**.
    - Name: users-datagen-lab
    - Partitions: 3
    - Create with defaults.
2. **Open the Connectors Catalog**
    - In the cluster left menu, click **Data integration** or **Connectors** (label varies).
    - You should see a catalog of source and sink connectors.
3. **Launch Datagen Source Connector**
    - Search for **Datagen** in the connector catalog.
    - Select **Datagen Source** (or similarly named quickstart connector).
    - Click **Configure**.
4. **Configure Datagen for Basic Demo**
    - Under configuration (UI may reflect fields from the quickstart docs):
        - **Name**: DatagenSource_users_lab
        - **Output topic**: users-datagen-lab
        - **Quickstart / Template**: USERS (or equivalent USERS dataset).
        - **Output format**: JSON_SR or JSON (if Schema Registry is enabled, JSON_SR is recommended).
        - **Tasks**: 1

- Provide the **Kafka API key/secret** for this cluster when prompted (follow the inline instructions or quickstart wizard).
- Click **Next / Continue** and then **Launch** the connector.

5. **Verify the Connector is Running**
    - Go to **Connectors** in the cluster and check DatagenSource_users_lab status is **Running**.
    - If it's Provisioning/Starting, wait until it becomes Running.

Part 5 – Observe the Stream in Message Browser

Goal: Show that topics are **append-only logs** and how events accumulate over time.

1. **Open users-datagen-lab Topic**
    - Go to **Topics** → click users-datagen-lab.
    - Open the **Messages / Message browser** tab.
2. **Watch Messages Arrive**
    - You should see a stream of user records, each with:
        - **Partition**.
        - **Offset**.
        - **Key** (if configured).
        - **Value** with fields like name, email, etc. (depends on USERS template).
    - Note how new messages appear automatically as the Datagen connector produces data.
3. **Explore Filters and Display Options**
    - Use filter options (e.g., filter by partition or value field) if available.
    - Increase the **max results** or adjust time range to show more history.

Part 6 – Consumer Groups and Lag in the UI

Goal: Connect the idea of **consumer groups** and **lag** to what Confluent Cloud displays.

For this part, you'll need at least one consumer application reading from users-datagen-lab with a specific **group id** (e.g., using Confluent CLI or a sample client). The observation itself is through the UI.

1. **Run Any Consumer (Outside the UI)**
    - (High level only, no deep code here.) Use one of:
        - A sample CLI consumer from Confluent's quickstart, configured with:
            - **Bootstrap**: your cluster's SASL_SSL://...:9092
            - **SASL** credentials (API key/secret)
            - **Group id**: users-lab-group.
    - Let it run for a few minutes so it reads from users-datagen-lab.
2. **Open Consumer Lag Page**
    - In your cluster's left navigation, open **Consumers**, **Consumer lag**, or **Client** tab (the name can vary with UI updates).
    - Look for consumer groups; find users-lab-group.
3. **Interpret Consumer Lag**
    - For users-lab-group, observe:
        - **Current lag** (messages behind producers).
        - **Records consumed per second**, if shown.
        - Partition-level lag breakdown.
    - Note that lag values may update with some delay (metrics latency) and may not show for inactive groups.
4. **Experiment**
    - Stop the consumer and watch lag increase as Datagen continues to produce.
    - Restart the consumer and see lag decrease as it catches up.

Part 7 – (Optional) Schemas and Schema Registry View

If your cluster has **Schema Registry** enabled and you chose JSON_SR as value format for Datagen, you can also illustrate schemas.

1. **Open Schema Subject for the Topic**
    - In the cluster navigation, click **Schemas**.
    - Look for a subject related to users-datagen-lab (e.g., users-datagen-lab-value).
2. **Inspect the Schema**
    - Open the subject and examine fields (e.g., name, email, etc.).
    - Discuss:
        - How producers validate messages against schemas.
        - How consumers use schemas for deserialization.
3. **Relate to Message Browser**
    - Go back to **Message browser** for users-datagen-lab and observe that messages deserialize according to the registered schema.